

Systematic review of sustainable food finance

Abstract

Despite rising global food production, the food system is fundamentally unsustainable, characterised by waste, unequal access to healthy diets, and a coexistence of overnutrition and hunger. Addressing these challenges demands systemic transformation, yet the role of capital markets in enabling or obstructing such transformation remains poorly understood. We conduct a systematic review of the academic literature on finance and food systems, combining structural topic modelling with qualitative analysis. The study reveals a narrow body of scholarship split across two largely distinct and epistemologically divergent paradigms: an optimisation-oriented perspective seeks to align capital markets with sustainability through better data and incentives, and a structural-critical literature, rooted in political economy, that views the institutional architecture of finance as a driver of food system unsustainability. Because these two traditions emerge from different theoretical starting points, they ask different questions and rarely engage one another's findings. We identify three consequences of this divide - functional, material, and methodological - that together produce lock-ins that stall cumulative progress. Rather than seeking to reconcile these traditions, we argue for a third position that takes the structural critique seriously while remaining open to the possibility that capital markets, under specific institutional conditions, could contribute to food system sustainability

1. Introduction

The global agri-food system has changed substantially over the last half century. Today, a few large agri-food corporations control the majority of the food supply chain, from seeds to retail, and through fertilizers, manufacturing and processing and distribution (Howard, 2021). The industrialized agri-food system is often celebrated for its productive capacity, but creates many negative externalities to society. Industrial agriculture, which is highly mechanized and heavily reliant on fossil fuels, chemical pesticides and fertilizers, is responsible for up to a third of global greenhouse gas emissions (Menegat et al., 2022) alongside significant biodiversity loss and water use (Crippa et al., 2021; Lawrence & Friel, 2019; Robinson et al., 2024).

The current agri-food system is also rife with contradictions; while it produces more food than ever before, over 40% of food is lost or wasted along the supply chain (Goodwin & Lipinski, 2024). Uneven access and distribution has led to a scenario where there is an overabundance of relatively cheap ultraprocessed food, while healthy and ecologically friendly food has become unaffordable to many (Swinburn et al., 2019; Willett et al., 2019). While well over a billion people are overfed, approximately 700 million people suffer from hunger and malnutrition (UN & WHO, 2025). These urgent and interlinked challenges have led to increased calls for food system transformation. While the urgency of food system transformation is widely recognized, there remains deep contestation over which direction and through which mechanisms it should proceed. Transformation at this scale demands more than policy adjustments; critical scholars suggest we dismantle existing power structures and reorient the capital flows that currently prop the existing regime.

This paper focuses specifically on *capital markets*, which we define as the institutions, instruments, and actors involved in allocating investment capital to food and agricultural systems through debt and equity markets (**CITE HERE**). This includes publicly listed agri-food firms, institutional investors, credit markets, and the regulatory structures that govern their interactions. We adopt this scope because capital markets represent one of the principal sites at which power over food systems is concentrated; decisions by asset owners and listed corporations shape production systems, land use, and supply chains globally.

The policy arena increasingly views capital markets as “a game changer to achieve food systems transformation” (“Finance for Food Systems Transformation,” 2023). Global food governance bodies propose that finance can be leveraged to bring about wide scale, sustainable outcomes. The International Food Policy Research Institute has estimated

that to meet climate change mitigation and adaptation targets, as well as other United Nations Sustainable Development Goals, food systems would require financing of \$USD350bn per year until 2030. Public financial resources alone would not suffice and thus private finance, including capital markets, would be heavily relied upon (“Finance for Food Systems Transformation,” 2023).

A report from the United Nations Environment Programme argues that “financial institutions with significant portfolio exposure to the agrifood sector [...] can influence clients and supplies across value chains to improve their policies and practices, demand accurate quantitative monitoring and reporting from investees and drive financial flows towards more sustainable food systems” (UNEP, 2023). Moreover, the 2021 United Nations Food Summit developed a comprehensive Food Finance Architecture, laying out how financial actors can “shift capital out of a high-carbon, unequal, extractive food system and into nature-positive, inclusive, climate-smart, circular business which create value for people, planet, and the economy” (Blended Finance Taskforce, 2021).

In March 2025, Access to Nutrition Initiative (ATNi), the Global Alliance for Improved Nutrition (GAIN), and the Paris Peace Forum presented The Paris Declaration on Business & Nutrition 2030. The Declaration sets a shared vision of an improved “Nutrition Economy” and calls on: investors and financial institutions to scale up concessional lending and private funds to drive healthier diet outcomes; institutional investors to demand change by calling for nutrition to be embedded into sustainability reporting and environmental, social and governance (ESG) investing practices; and the private sector to take responsibility of nutritious options, and shift marketing strategies away from unhealthy foods (ATNi, 2025).

These developments are reminiscent of the ways in which financial actors have been targeted as high-leverage intervention points to limit the expansion of the fossil fuel industry (Dordi et al., 2023). From sustainable investing standards to shareholder activism, in many ways, the green energy transition provides a blueprint for those seeking to influence change in the food system. ESG investing in the food system can target a range of issues that include soil health, land use practices, animal welfare, fair labour standards and packaging. On the philanthropic side, the TransCap Initiative, backed by the Walton Family and Rockefeller Foundations aims to build coordinated portfolios of investments to shift systems towards more regenerative practices in the Midwestern United States, the country’s agricultural heartland. Activist groups are beginning to target large agri-food corporations they deem responsible for the multiple ills of the industrial food system. A coalition of non-governmental organizations (NGOs) led by Friends of the Earth and the Global Alliance for the Future of Food are developing shareholder activism strategies to pressure big agri-food corporations into

abandoning “dirty” practices (**CITE HERE**). Yet, as this review suggests, food systems may require fundamentally distinct strategies than those developed for fossil fuels.

Momentum is building around ways to leverage finance capital to transform the food system, but the academic literature does not reflect the growing activity in this space. Two bodies of scholarship inform this intersection of finance and the transition towards more sustainable food systems, but neither directly addresses the role of capital markets as leverage points for sustainable food transformation.

The sustainability transition literature recognises finance as consequential (Mazzucato & Semieniuk, 2018), yet rarely examines food systems specifically. A few articles have been published recently on the role of financial actors in supporting sustainable food transitions (Robinson et al., 2020, 2022, 2024; Stephens, 2021b, 2021a). While they provide valuable insights into investor perspectives on financing sustainable food transitions, they are focused on specific country or regional case studies and do not engage with the large agri-food corporations that dominate the industrialized food system.

Since the food and financial crises of 2007–2008, many researchers have studied the financialization of the food system, and its impact on social and environmental outcomes. For instance, studies have been conducted on the ways in which the rise of speculative investment in agricultural commodity markets influences food insecurity and harms farmer livelihoods (**cite**); the rise of shareholder value and how it shapes agri-food companies’ behaviours to prioritize short-term profit over other more sustainable, long-term goals; and the ways in which speculative investment in farmland has reshaped landscapes towards larger scale, more industrialized agricultural practices (Fairbairn, 2021). This scholarship identifies financialization itself as a primary driver of unsustainable outcomes. Its analytical strength lies in asking whether, or how, financial capital might be redirected toward more sustainable outcomes.

Scholarship at this intersection exists, but is split between traditions that rarely engage with one another. Mazzucato and Semieniuk (2018) argued that finance is not neutral and we take this approach as well: the types of financial tools available and the mentality of investors has a profound impact on the directionality of a transition. Given the urgent need to transform food systems, and the increasing structural power of capital markets, we must reassess their role. We therefore take as our starting point, two conditions. One, that food systems required urgent change and two that capital market actors have become too powerful to leave outside the analysis. This requires us to move past the respective blindspots of both the ‘optimization’ and ‘critical’ literatures; beyond the assumption that better metrics alone will suffice, and beyond the assumption that financial capital can only entrench harm. We undertake this review to

explore whether a ‘third way’ exists; one that acknowledges the urgency of change while rigorously addressing the structural flaws inherent in current financial logic. This question, we suggest, is also the one that civil society actors seeking to mobilise finance for food system change are beginning to pose in practice.

In this paper, we ask: What does the structure of academic literature at the intersection of capital markets and food systems reveal about the deeper barriers to sustainable food finance? We systematically review 71 peer-reviewed articles using the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) protocol and Structural Topic Modelling (STM), combining quantitative analysis with qualitative interpretation, to map dominant themes, trace their evolution, and identify the gaps that explain why capital markets have so far failed to drive meaningful food system transformation.

We find, through our analysis, a field that is narrow, fragmented, and operationally disconnected from the biological complexity it seeks to manage. On one side, an optimization perspective operates broadly within the existing regime of capital markets. Namely, this literature asks how environmental, social, and governance (ESG) data, sustainability disclosure, and responsible investment practices can better align financial incentives with food system sustainability. On the other side, a structural perspective, rooted in political economy, asserting that the very design of capital markets - shareholder primacy, short termism, and commodification - are the drivers of food system unsustainability. These two literatures speak largely in parallel, indicative of the clash between a fundamental misalignment between the short term, liquidity-driven logic of financial markets and the spatial, biological, and temporal complexity of ecological systems. Consequently, we lack a shared evaluative framework for determining whether capital markets are contributing to food system sustainability or merely performative while reinforcing unsustainable regimes.

2. Methods

To systematically map the literature at the intersection of food systems, finance, and sustainability, we apply a STM approach to identify dominant themes, trace their evolution, and develop a new conceptual framework for understanding how capital markets interact with food system sustainability. Guided by the PRISMA protocol, we proceed in three stages: 1) identifying and screening a sample of relevant peer-reviewed articles; 2) conducting a quantitative STM analysis of the resulting corpus; and 3) presenting a qualitative interpretation of the themes generated by the STM. Each stage is described in turn below.

2.1 Data collection and screening

A systematic search of the literature was first conducted to identify peer-reviewed articles examining the intersection of finance and food systems. Our search strategy employed title-based queries (presented in figure 1) across two leading academic databases, Scopus and the Web of Science, that explicitly connect finance and food systems, with an added keyword for sustainability to ensure relevance to our research goal. The initial search yielded a total of 1,637 records.

Imported data was cleaned using the *Tidyverse* collection of packages on R. Titles and journal names were standardized to correct formatting inconsistencies, such as the use of ampersands and irregular capitalisation. Keywords were also consolidated by merging author-provided keywords (ID) with database-assigned keywords (DE) into a unified field (ID_DE).

Following the PRISMA approach, commonly applied in both food studies (Elsner et al., 2023) and finance (Dedousi et al., 2025; Dordi et al., 2023; Galletta et al., 2024), we applied a multi-stage deduplication process to remove exact and near-duplicate articles across both databases. We first removed entries with identical references and citation counts. We next removed entries with identical title, journal, and publication year, to retain the record with the highest citation count. We finally removed near-duplicates, based on the first eight words of the abstract, to capture similar studies with slight title variations. This was important as Scopus and the Web of Science databases likely yielded duplicates. This exclusionary approach resulted in a final sample of 1,113 articles starting from an initial 1628 articles (801 articles from Scopus and 827 from Web of Science respectively).

Finally, the remaining articles underwent a manual relevance screening by three co-authors to ensure their alignment with the core research aim, namely the role of capital markets in food systems. To ensure interrater reliability, articles were excluded if at least one of the three reviewers deemed them outside scope. Inter-rater agreement was assessed across the full sample. All three reviewers had full consensus (all three reviewers agreed that a paper was out of scope) on 64% of articles, and majority agreement (two of three reviewers agreed that a paper was out of scope) on 78%. To ensure no relevant articles were inadvertently excluded, a fourth reviewer subsequently assessed all discarded articles, confirming or modifying the final selection. After this process, the number of papers in the sample totaled 71.

During this iterative review process, several distinct thematic clusters emerged that, while related, fell outside our defined scope of capital markets. Notably, we excluded papers that centered on: 1) farm-level economics without a link to external capital markets; 2) financial inclusion topics such as microcredit for poverty-oriented lending; 3) public or development finance (e.g., government subsidies, World Bank funding) that did not engage with private investment; 4) papers with a purely macroeconomic focus, 5) those centered on non-food sectors (e.g., forestry for timber, solar farms), and 5) those examining alternative finance models like local crowdfunding. These domains operate according to fundamentally different logics to

capital markets - centring for example, on security, poverty alleviation, or community resilience rather than risk-adjusted returns. Thus their inclusion would have obscured the specific mechanisms through which capital markets interact with food systems.

This process resulted in a final, validated sample of 71 peer-reviewed articles for thematic analysis. Although the final sample is modest in size, this is in itself a substantive finding: reflecting the genuine scarcity of peer-reviewed scholarship at the specific intersection of capital markets and food system sustainability.

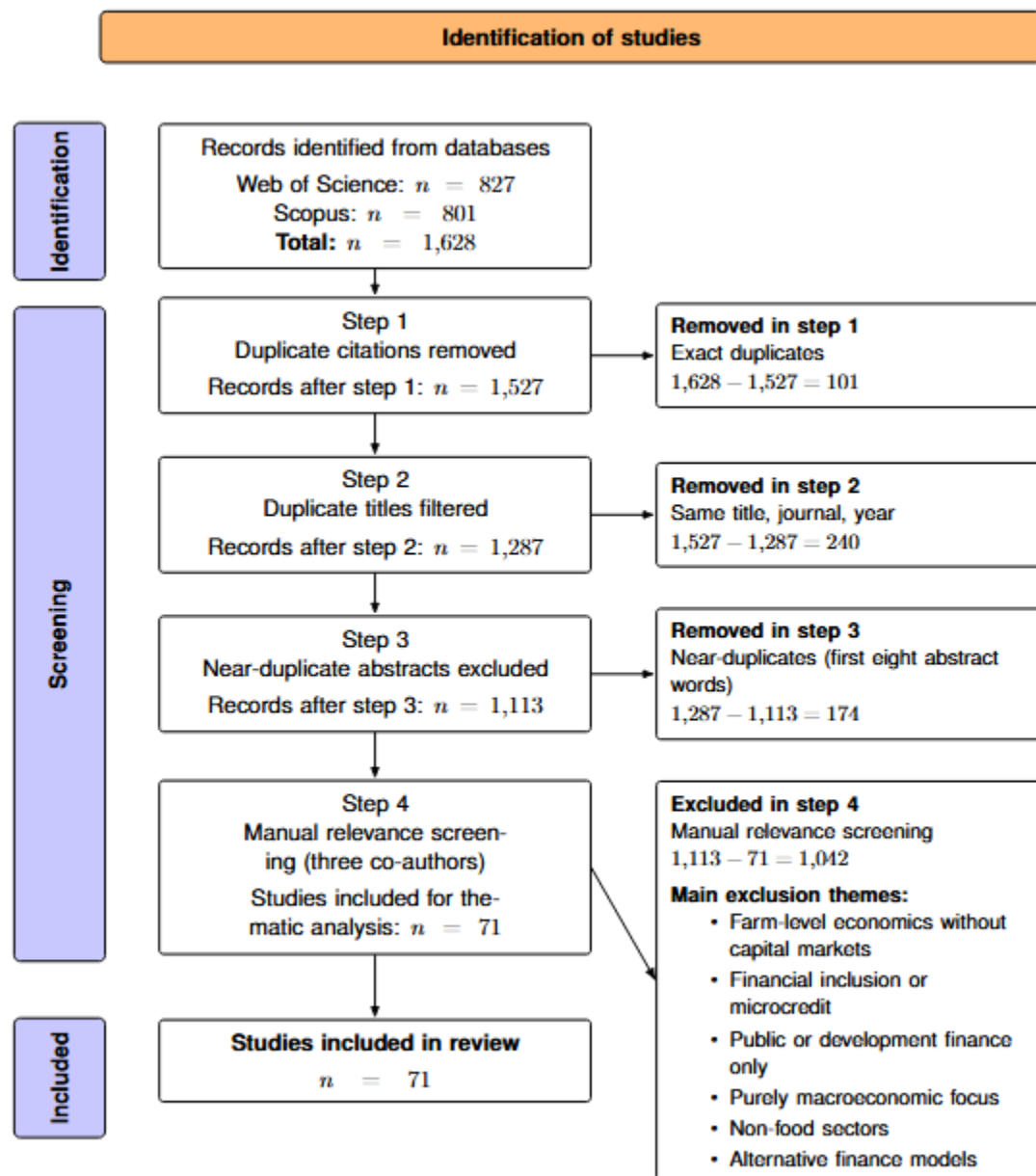


Figure 1 Search query and exclusionary screening

2.2 Structural Topic Modeling

We used Structural Topic Modeling (STM) to identify the latent thematic structure of our sample corpus. Unlike conventional content analysis, STM is unsupervised (themes are inferred algorithmically from the text rather than imposed by the researcher), making it well suited for exploratory systematic reviews (Blei et al., 2003; Roberts et al., 2014). STM assumes that each document in the corpus is a mixture of topics and each topic represented as a distribution over words (Blei et al., 2003; Blei & Lafferty, 2007; Roberts et al., 2016). In practice, an article on sustainable food finance might draw on a 'green finance' topic (assigning high probability to words like 'SDG', 'blended finance', 'decentralised') and a 'financialization' topic (assigning high probability to words like 'shareholder', 'ETF', 'land grabbing') simultaneously, with the model estimating the proportion of each.

A key advantage of STM over related approaches such as Latent Dirichlet Allocation (LDA) is that it does not assume a uniform prior distribution over topic proportions. Instead, it places a logistic-normal prior on document topic proportions, which allows for more flexibility in representing how topics might vary across documents, particularly in relation to external covariates or document-specific characteristics. In our case we use the date of publication of each article in our corpus as a covariate to help us identify how different contexts can influence a topics (Blei & Lafferty, 2007; Li et al., 2021).

By analysing the topics STM infers from the literature alongside their co-occurrence patterns (Jung et al., 2023; Mishra & Jiang, 2022), we are able to examine how themes such as sustainable investment, supply chain governance, and environmental performance are relationally structured and to generate new theoretical insights about how financial and food system discourses intersect without imposing predefined categories.

Prior to modelling, the keyword corpus underwent machine-led pre-processing, in four steps, to optimise it for STM. First, the selected keyword corpus underwent pre-processing to optimize it for the Structural Topic Modeling (STM) model. The keyword strings were tokenized by splitting the semi-colon delimited terms. Next, common stop words (e.g., "the", "and", "is") and custom list of stop words from our initial search query (e.g., "sustain," "finan," "invest") were removed to prevent the model from simply identifying topics based on the original search criteria. Finally, all remaining words were reduced to their lexical root using Porter's stemming algorithm to group related terms (e.g., "finance," "financial").

Secondly, to ensure the validity of our findings, we undertook a multi-stage process to select the most effective model. Initially, three separate text sources were considered: (1) titles alone, (2) abstracts alone, and (3) keywords. Models were assigned a confidence score, based on the proportion of articles with a fit (gamma) of 0.8 or higher (which is standard in the literature, see Hamm & Odrowski (2021)). Following low confidence scores among titles and abstracts (0% and 62.4% respectively) versus the keyword model (87.7%), titles and abstracts were subsequently combined into a single text. The keyword-based model assigned topics with substantially higher confidence (90%) compared to the combined title and abstract model

(48.7%). Consequently, we selected the author- and indexer-provided keywords as the basis for our final thematic analysis.

Next, we determined the number of topics (K) using a two-step process. First, we evaluated the model fit for each $K = \{5, 10, 15, 20, 25, 30\}$ using the held-out likelihood and conducting residual analysis as these are useful indicators of a model's fit (Taddy, 2012; Wallach et al., 2009). Then, given that there is no single "correct" value of K , we combined these diagnostic tools with qualitative judgement to balance interpretability with the statistical goodness of fit (Roberts et al., 2014). Following this two-step process, we determined the optimal number of topics to be $K = 5$.

Finally, we rely on three primary STM outputs in the analysis and reporting. First, the average topic proportions across the entire corpus indicate the share of words assigned to each topic and, as this provides a measure of the overall salience of each theme in the dataset. Secondly, the highest probability words associated with each particular topic identify the most representative terms of a topic, as these are an intuitive way to interpret the substantive meaning of the topic. Lastly, we also use the FREX metric, which balances frequency and exclusivity. Intuitively, if a word with high probability under one topic has low probabilities under the rest, then we say that that word is exclusive (Roberts et al., 2016, p. 994).

2.3 Qualitative Synthesis

The STM analysis identifies the thematic structure of the corpus but does not, on its own, explain how those themes relate to one another or how the scholarly debate within each has evolved. To address this, we complement the STM output with a qualitative synthesis of the articles within each topic cluster. This interpretive aspect allows us to delve deeper into the substantive content of each theme (the theoretical frameworks, empirical contexts, and conclusions drawn) and to identify the nuances that a purely quantitative approach would miss.

For each of the five topics, we reviewed the ten highest-cited articles in depth, treating citation counts as a proxy for scholarly influence within the field. We read these articles not to classify them into discrete categories (since STM assigns each article a probability distribution across all topics) but to build an inductively grounded account of how each theme has developed and where it currently stands. Our interpretation was further informed by the full corpus, with less-cited articles drawn upon where they offered contrasting perspectives or addressed gaps in the dominant literature.

Overall, our methodology has some notable limitations. The title-based search strategy, while precise, may have excluded relevant papers that address the finance-food intersection without using our specific terms in their titles. The review is limited to articles indexed in Scopus and Web of Science, and does not capture books, policy reports, or grey literature. Finally, while STM provides a data-driven thematic structure, the interpretation and labelling of topics is inevitably shaped by our collective analytical perspective.

Results

The final corpus of 71 peer-reviewed articles spans three decades of scholarship, ranging from 1994 to 2025. Research output remained negligible until the mid 2000s with a first inflection point emerging after the 2008 global financial crises - when interest in large-scale agricultural investment, land grabbing, and food security intensified. A second and more pronounced surge emerged from approximately 2020 onwards, driven in part by the convergence of the EU Green Deal, corporate ESG momentum, and growing interest in climate finance. This acceleration is itself an interesting finding; that the literature is not only sparse but quite recent, with the majority of relevant scholarship produced in the last five years.

3.1 Structural Topic Modeling

Using STM, we identified five distinct topics within the corpus. We report three complementary outputs for each; the FREX scores (Figure 1), which identify words that are both frequent and exclusive to a given topic (thus capturing what makes each theme analytically distinct); the term-topic probability scores (β) (Figure 2), which identify the most representative words overall; and the expected topic proportions (Figure 3), which indicate each theme's relative salience in the corpus.

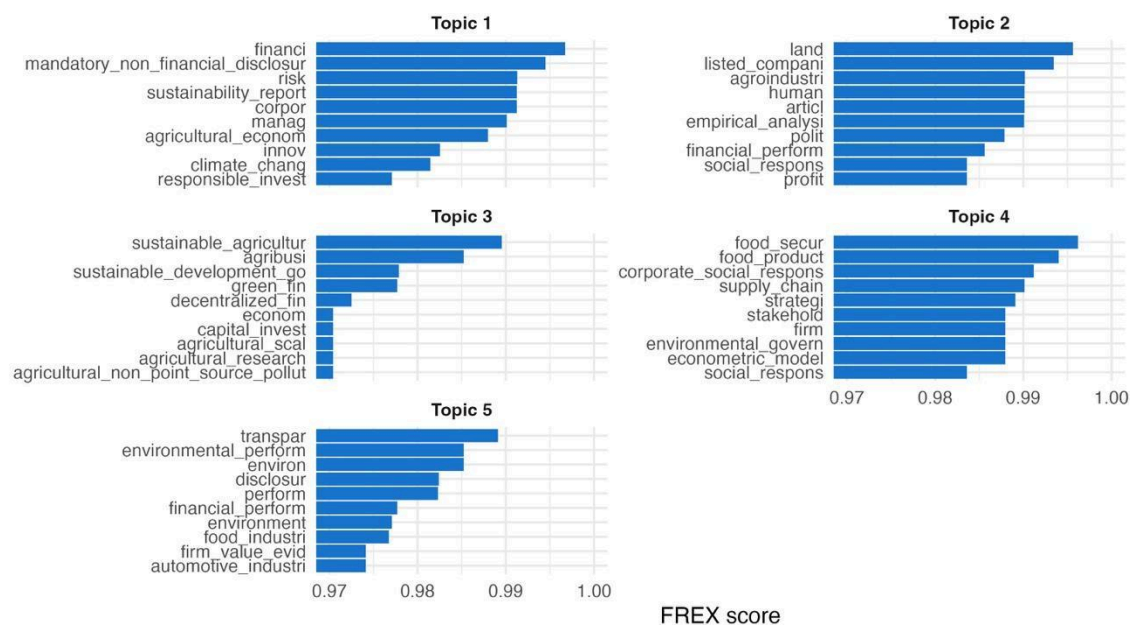


Figure 1 Top Words per Topic (FREX)

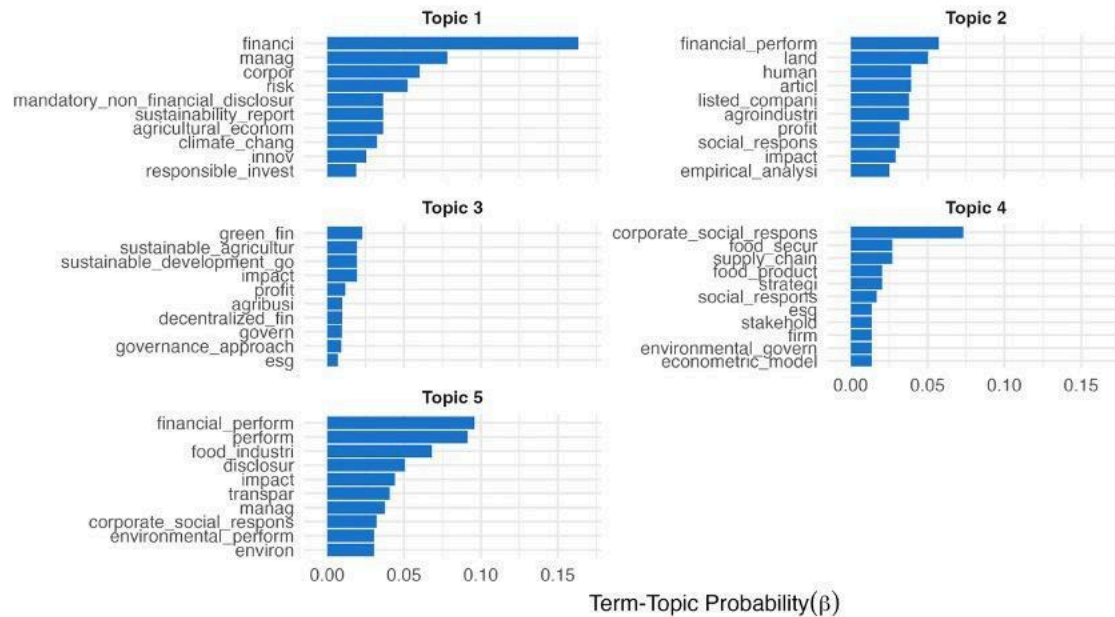


Figure 2 Top Words per topic (Probability)

Topic 1 - *Financialization and Risk Management* - is predicated by terms like financialization, non-financial disclosures, and corporate management. While the least 'prevalent' theme in the corpus (Figure 3), the terms in this topic have among the highest term concentration (Figure 2). Specifically, financialization has a notably high beta score, indicating that papers within this cluster treat it as a central organising concept. This suggests a small but focused body of scholarship on the structural consequences of financial logics in agri-food systems.

Topic 2 - *Agroindustry: Financialization and CSR* - is concerned with an empirical link between social responsibility, financial performance, and profit in the agri-food contexts, often drawing on large datasets of listed firms and econometric methods. It is characterized by terms like listed companies, empirical analysis, and profit, suggesting that this corpus evaluates conventional financial logics that views sustainability efforts in a manner that aligns with neoliberal framings.

Topic 3 - *Sustainable Agriculture and Green Finance* - is by far the most prevalent theme in the corpus (Figure 3), anchored by terms including green finance, sustainable development goals, decentralised finance, and agricultural scale. The breadth of this topic is reflected in its comparatively lower term-topic probabilities, with no single term dominating; suggesting a diffuse but voluminous conversation about the role of finance in promoting sustainable agricultural development, often framed around SDG targets and blended finance instruments.

Topic 4 - *Food Security and Supply Chains* - is characterised by terms like corporate social responsibility, food security, supply chains, and food production. The presence of terms related to environmental governance and stakeholder engagement reflects scholarship concerned with multi-actor governance across food supply chains, with a particular emphasis on how listed corporations manage and disclose their sustainability commitments in relation to food security outcomes.

Finally, Topic 5 - *Environmental Performance and Transparency* - is driven by terms like transparency, disclosure, financial performance, and firm value. This theme sits at the intersection of corporate accountability and market signalling, examining whether and how environmental reporting practices translate into financial outcomes, and whether disclosure functions as genuine risk management or performative compliance.

The uneven topic prevalence across the corpus in Figure 3 below is also analytically significant. Topic 3 (Sustainable Agriculture and Green Finance) accounts for the largest share of the literature and reflects the strong influence of green finance and SDG discourse in shaping the field. Topics 4 and 5 occupy a middle ground, together suggesting considerable scholarly attention to corporate disclosure and supply chain governance. Topics 1 and 2 are the least prevalent, yet Topic 1 exhibits the highest conceptual concentration, with financialization functioning as a core grounding for the papers that do engage with it. This pattern is indicative of one of our central arguments - while the structural critique of financial markets is present within this corpus, it constitutes a relatively small portion of the corpus relative to the optimization-oriented mainstream.

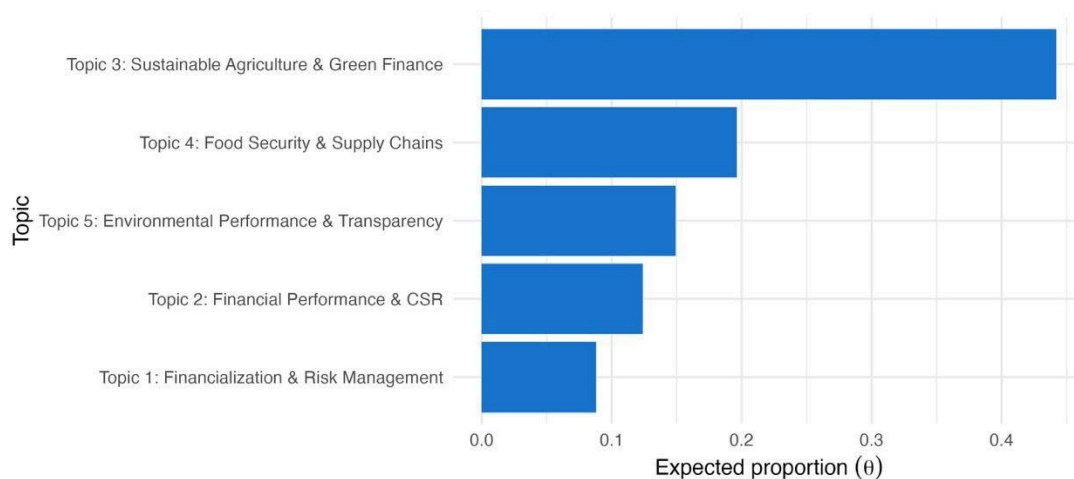


Figure 3. Topic Prevalence

3.2 Qualitative synthesis of the STM output

To move beyond the statistical structure of the corpus, we complement the STM output with a qualitative synthesis of the ten most-cited articles within each topic cluster. Our readings reveal an area of inquiry characterised by two foundationally distinct scholarly camps. Rather than a unified field, we observe competing framings, uneven empirical foundations, and a persistent tension between optimisation-oriented and structurally critical approaches. Synthesizing these two separate bodies of literature allows us to explicitly contrast the potential and limitations of financial mechanisms in driving agricultural sustainability.

Topic 1: Financialization and Risk Management

The relationship between environmental disclosure practices and firm financial performance represents a narrower but theoretically significant strand of research. Drawing on stakeholder theory and legitimacy theory from the corporate governance literature, scholars have investigated whether non-financial environmental disclosures are correlated with improved financial outcomes. Highly cited studies, such as those by Nirino et al. (2019) and Vitale et al. (2023), have identified positive relationships between firm-level environmental disclosure and financial performance, suggesting that transparency regarding environmental impacts may signal effective risk management and long-term value creation to investors. However, critical perspectives on corporate financial actors in the agri-food system complicate this narrative once again. Using the phenomenon of financialization, here defined as the increasing dominance of financial logics not just in farmland but in agri-food corporations, Clapp (2019) documents the concentration of ownership among large blockholders which pool equity capital passively through Exchange Traded Funds (ETFs), restricting competition. While agri-food sector ETFs increasingly disclose their ESG risks to investors, questions remain regarding whether such disclosure mechanisms are sufficient to counteract the market-restricting behaviours of concentrated ownership structures. This tension between transparency, disclosure, and substantive change represents an essential limitation of disclosure-focused approaches. Additionally, this tension raises a more fundamental question: whether financial actors meaningfully utilize environmental information in their decision-making processes, or whether disclosure practices function primarily as performative.

Topic 2: Agroindustry: Financialization and CSR

A rapidly growing strand of literature that emerged in response to the keywords found in the previous section (FREX and PROB keywords) such as “land” and “agroindustrial” examines the phenomenon of financialization in the context of agricultural land as an asset class. Work by Clapp and Isakson (2018) as well as Ceddia (2020) trace the emergence of farmland itself as a financial investment vehicle and its implications for sustainable agricultural transitions. These papers argue that when farms are managed to maximize investor returns rather than promote sustainable practices, the environmental consequences can be severe. For instance, Ceddia's (2020) empirical analysis of Brazil demonstrates that rising farmland financialization drives deforestation; specifically, a one percent increase in wealth concentration among high-net-worth individuals correlates with a 2.4 to 10% percent expansion in flex-crop cultivation. This research reveals how financial capital flows, particularly foreign direct investment concentrated in Latin America and Southeast Asia, intensify land-use pressures and environmental degradation, even as firms engage in green finance initiatives and ESG disclosure practices.

Topic 3: Sustainable Agriculture and Green Finance

The largest body of literature examines the potential of “green finance” instruments to promote sustainable agricultural development. More critical approaches note a fundamental debate over the role of private finance in development trajectories.

Empirical studies by Gao et al. (2022), Jiménez-Aliaga et al. (2022), Gernego et al. (2022), and Taridala et al. (2023) investigate diverse green financial mechanisms across different

geographic contexts, particularly in the Global South and generally identify positive associations between such instruments and sustainable development metrics. Other papers in this theme interrogate the use of specific financial instruments in enabling sustainability transitions, highlighting the opportunities presented by blended finance structures to mobilize private capital on a large scale for sustainability transitions. For instance, Havemann et al. (2020) discuss the variety of blended finance architectures that are available, ranging from concessional loans and direct grants to equity and debt instruments. At the same time, they acknowledge substantial implementation challenges, including data quality deficits, concerns about counterparty creditworthiness, and limited market liquidity. Despite conceding such limitations, this kind of framing broadly positions financial innovation as essential to meeting development goals, given the scale of capital requirements.

Critical scholars examining financialization dynamics in agri-food systems complicate this optimistic narrative. For instance, Martin and Clapp (2015) and Clapp and Isakson (2023) emphasize the historical and continuing role of the public sector in stabilizing agricultural investment flows. This strand of scholarship argues that private financial markets, driven by shareholder value maximization, are inherently volatile and insufficient for sustaining large-scale agricultural investment amid environmental uncertainty. Instead, public sector actors and intervention, through both direct finance provision and support for agricultural research, remain critical to achieving genuine sustainability transitions. The tension between these framings reflects broader questions about whether market-based mechanisms prioritize sustainability or can be restructured to do so. Alternatively, is more fundamental institutional reform necessary? Setting aside ecological justice aims, a question that now arises is the relationship between environmental disclosure and economic performance. What value do green finance instruments and/or ESG disclosure practices bring in terms of financial performance?

Topic 4: Food Security and Supply Chains

Emerging scholarship emphasizes the inadequacy of generic financial assessment frameworks for agricultural contexts, advocating instead for sector-specific approaches to sustainability evaluation, as reflected in the FREX terms “food production” and “supply chains”. Well-cited papers such as Rueda et al. (2017) develop an analytical framework for understanding corporate decision-making regarding sustainability strategy adoption, encompassing source labelling, ecosystem-sensitive sourcing, and livelihood improvements. Importantly, their work highlights how agricultural production has evolved from isolated operations to integrated positions within regional and global networks, thereby creating new transparency imperatives. A related theme concerns the inadequacy of generic financial rating systems for agricultural contexts. Gomes et al. (2023) argue that different food sectors, such as dairy manufacturing versus processing companies, exhibit distinct environmental-financial relationships that conventional rating systems, like Moody’s, fail to capture, resulting in inaccurate risk assessments. Similarly, Robinson et al. (2024) examine how institutional investors might catalyze food system sustainability in the Australian context, concluding that existing ESG, divestment, and engagement tools developed for other sectors cannot be directly transferred to food production contexts without substantial adaptation. Even more tellingly, research from

China provides cautionary evidence that green finance initiatives, while intended to promote sustainability, can simultaneously reduce agricultural production, thereby diverting funds away from food production (Lee & Song, 2024). The crucial discussion here is, therefore, whether existing metrics in the field fulfil their intended purpose, which in turn, begs the question of their impact. After all, managerial responses to ESG and financialization, spanning from investment mechanisms to corporate disclosures, rest on not just whether the information and frameworks available are adequate and credible but whether they are 'used' and how.

Topic 5: Environmental Performance and Transparency

Given the inadequacy of existing metrics, our last category of small but analytically significant literature examines how financial actors actually respond to and utilize corporate environmental reporting, interrogating whether disclosure frameworks meaningfully inform decision-making. The papers discuss the strategies and communication tactics that firms use to signal their alignment between sustainability goals and financial objectives to their investors and stakeholders, ranging from supply chain instruments to formal ESG reporting, with the twin objectives of enhancing investor perceptions and reducing the cost of capital. More importantly, papers falling within this theme explore how financial actors respond to and interpret such environmental reporting by agricultural corporations. Unsurprisingly, the scholarship remains divided. Partalidou et al. (2019) find positive associations between environmental performance indices and economic returns, potentially supporting arguments that environmental stewardship represents effective risk minimization or long-term value creation. However, a qualitative analysis of investor sentiment expressed in New Zealand media (Perera, 2024) reveals considerable heterogeneity in investor attitudes toward environmental disclosure. Investor responses thus cannot be reduced to simple positive or negative correlations. This emerging recognition of behavioural complexity indicates a crucial gap: understanding the psychological and institutional drivers underlying investor interpretation of environmental information in agricultural contexts remains underdeveloped.

Overall, this broad synthesis suggests that traditional tools and frameworks of green financing and ESG disclosure, as outlined by papers such as Nirino et al. (2019), Vitale et al. (2023), and Partalidou et al. (2020), do not adequately capture the material reality of agricultural production, leading to potentially suboptimal decisions that managerial efforts claim to solve. This echoes arguments of 'agricultural exceptionalism,' suggesting that the biophysical and social specificity of food systems resists standard financial capitalization, leading to potentially suboptimal decisions. The literature on green financing and ESG disclosures collectively demonstrates that financial mechanisms, whether disclosure systems, blended finance instruments, or ESG frameworks, operate within broader structural constraints and exhibit contradictory effects.

Across these five topics, a consistent disconnect emerges. The financial mechanisms evaluated, whether disclosure systems, blended finance, or ESG frameworks, are frequently misaligned with the socio-ecological contexts they attempt to measure. Where scholarship engages the biophysical specifics of food production (Topics 4 and 5), it finds generic financial tools inadequate; where it does not (Topic 3), it tends to report finance and sustainability as

compatible. Taken together, the patterns we identified suggest that the limitations of sustainable food finance are not solely a matter of insufficient capital, but reflect a deeper, structural incompatibility between standardized financial tools and the heterogeneous realities of food production. These tensions form the basis for the theoretical framework developed in the following section.

4. Discussion

Our review reveals a fragmented understanding of the role of capital markets in global food systems, capturing two coherent traditions proceeding from incompatible epistemological and political starting points. The interaction between capital markets and food systems is not only a technical challenge of closing the investment gap as some would argue. The optimisation topics (3 and 5) and the structural topics (1 and 2) diverge in what they treat as the object of analysis: the former examine the financial logic driven by efficiency and quarterly returns, as a tool whose calibration can be improved; and the latter examine finance as an institution whose logic produces the outcomes in question.

On this basis, we make three contributions. First, that the current literature on food, finance, and sustainability barely address capital markets. Of the 1,637 articles identified through the selection process, just 71 articles speak to the role of capital markets specifically, less than 5 percent of the total. This is critical given the influence capital markets have on global agrifood systems and their perceived role in driving sustainability transitions. This scarcity of literature examining the role of institutional investors, listed companies, and capital markets highlights a significant gap; while policymakers are banking on private finance to transform the food system, the academic understanding of how these specific actors operate within the agri-food sector remains thin and largely fragmented. Second, we identify three structural tensions that run through this literature and, when taken together, produce what we term an ‘informational lock-in’: the condition in which the absence of actionable sector-specific metrics prevents capital reallocation even when the political will to act exists. We argue, however, that this lock-in is not the skeleton key that explains everything but only a part of the story that operates alongside the structural dynamics identified by political-economy scholarship. Third, we propose that solving these tensions requires taking the structural critique seriously while remaining open to the possibility that capital markets, under specific institutional conditions can and must play a constructive role in food system transformation.

4.1 The Structural Barriers to Change

While there is interest for sustainable food systems transformation among capital markets and policy, these efforts remain disconnected from ground-level outcomes. We argue that this is symptomatic of deeper structural barriers, which we reveal across three core barriers below.

4.1.1 The Functional Barriers

First, a functional barrier concerning the ‘mechanism’ of change: there is no shared account of why capital fails to flow toward sustainable food systems. The results reveal a division between scholarship that views finance as a neutral tool to be optimized for sustainability (topics 3 and 5) and scholarship that views finance itself as a structural constraint (topics 1 and 2). This infers that there remain competing understandings of why market failures (such as environmental or health externalities) persist in food systems.

The dominant strand of literature, represented by Topics 3 and 5, approach the challenge through the lens of information asymmetry, asserting that capital misallocation stems from a lack of transparency; that better data will enable markets to achieve win-win outcomes. Aligning with a ‘green finance’ paradigm, studies within this cluster, such as those by Nirino et al. (2019) and Vitale et al. (2024), identify positive correlations between environmental disclosure, ESG, and financial performance, reinforcing the narrative that sustainability signals are effectively priced in by the market.

The critical body of literature (Topic 2) in contrast, argues that market failure is rather a structural challenge. Aligning with the financialization of food systems critique, studies like Clapp and Isakson (2018) contend that the very design of financial markets, shareholder primacy and short termism, is incompatible with sustainable agricultural transitions. Ceddia (2020) demonstrates, for example, that the prioritisation of shareholder returns correlates directly with flex-crop expansion and deforestation in Brazil. Similarly, Clapp (2019) shows how concentrated ownership via passive investment vehicles like exchange traded funds restrict competition and further entrench industrial agricultural models.

This division contradicts each other and consequently restricts collective response. While the optimization literature presupposes that the solution lies in refined metrics and transparency, the transformation literature warns that improving the quality of information is futile if the underlying structural incentives force firms to prioritize short-term extraction over long-term resilience.

4.1.2 The material barrier

Second, there is a material tension regarding the ‘tools of change’. We assert here that the use of sustainable finance is generic, built for energy, technology, and manufacturing, and applied rather clumsily to the food system. By doing so, the tools we use are inadequate to capture the heterogeneous biological reality of agri-food systems. What is required is sector-specific tools.

This mismatch emerges in the disconnect between scholarship on transparency (Topic 5) and sector specific supply chains (Topic 4), which emphasize that generic financial assessment frameworks are inadequate for agricultural contexts. Robinson et al. (2024), for example, argue that existing divestment, and engagement tools, which were developed for other sectors, cannot be directly transferred to food production contexts without adaptation. Unlike the fossil fuel sector or manufacturing, where carbon provides a relatively universal metric, food systems involve complex biological processes that can not be standardized under a single metric.

This lack of specificity leads to inaccurate risk assessments. Gomes et al. (2023) find that conventional rating systems fail to capture the diversity of financial influence across food sub-sectors, such as between dairy manufacturing and food processing. Relatedly, this risk varies by region; Rueda et al. (2017) assert that corporate decision-making must account for ecosystem-sensitive sourcing and livelihood improvements, which vary significantly across regional networks. When financial tools ignore these nuances in favor of standardization, they fail to capture the material risks and opportunities unique to food production.

Finally, this mismatch can drive counterproductive outcomes. Lee and Song (2024) find, in the context of China, that green finance initiatives, while intended to promote sustainability, can reduce agricultural production and divert funds away from the sector entirely. When financial mechanisms are not calibrated to the material reality of food production, they may incentivize capital to flee the complexity of agriculture rather than transform it.

This challenge extends beyond environmental metrics. The 2025 Paris Declaration on Business and Nutrition highlights that nutritional outcomes - a critical dimension of food system sustainability - remain almost entirely absent from existing ESG and investment frameworks, despite explicit calls for institutional investors to embed nutrition into sustainability reporting (ATNi, 2025). This further indicates how far generic financial frameworks are from capturing the material reality of the sector.

4.1.3 The Methodological Barrier

Finally, we note a fragmentation in the literature by a methodological divide. Quantitative studies, often utilizing large datasets scraped from financial terminals, generally identify positive associations between sustainability and financial performance, whereas qualitative and critical studies frequently uncover negative or complex outcomes. For instance, Partalidou et al. (2020) link environmental performance indices to economic returns, reinforcing the view that financial markets reward green behavior. In contrast, by interrogating the mechanisms behind these numbers reveal different conclusions, Clapp (2019) and Ceddia (2020) show how financial actors may have structural disincentives to prioritize sustainability. Further, Perera's (2024) analysis of investor sentiment reveals considerable heterogeneity in how environmental information is interpreted, challenging the assumption that markets rationally "price in" disclosure.

We posit that this methodological barrier may also emerge from what is observed. Quantitative scholarship tends to measure what is easily observable (disclosure scores and stock prices), while qualitative scholarship examines the structural drivers (power and land use). This disconnect implies that the "positive link" between finance and sustainability often cited in the literature may be due to data availability, rather than a reflection of on-the-ground reality.

Taken together, the functional, material, and methodological barriers form a mutually reinforcing system. This divergence obscures the informational lock-in at the heart of the problem. Because the dominant methodology validates generic ESG frameworks, it provides institutional cover for

a measurement infrastructure that is fundamentally misaligned with the biological reality of food production. Disclosure then becomes a form of symbolic compliance; while firms may satisfy reporting requirements for "sustainable development goals" or "transparency" - without sector-specific metrics that respect the temporal and spatial realities of agriculture, these disclosures fail to bridge the gap between financial capital and sustainability.

4.2 Mapping the informational lock-in

Sustainability transitions theory assert that stabilizing 'lock-in' mechanisms hinder transformative change. Typically, these barriers are categorised as technological lock-in (sunk costs in physical infrastructure), institutional lock-in (regulatory frameworks and norms), or behavioral lock-in (habits and routines) (Geels, 2011; Unruh, 2000). We identify a distinct fourth mechanism: informational lock-in, defined as a form of lock-in that occurs when the absence of decision-useful, sector-specific metrics prevents regime change. Even when technological alternatives exist, institutions support transition, and actors are motivated to change.

First, a clarification is essential here. This informational lock-in does not operate in isolation. Rather, it does so through the structural mechanisms identified by the critical scholarship such as concentrated ownerships (Clapp, 2019), shareholder primacy and short-termism (Clapp and Isakson, 2017), and the financialization of farmland as an asset class (Ceddia, 2020; Fairbairn, 2020). As such, we argue that these are not parallel explanations but interlocking ones: concentrated capital lacks the incentive and capacity to engage substantively with sector-specific metrics; quarterly reporting leads to the mispricing of long-term ecological indicators; and shareholder primacy leads to a preference of short-term returns over well-designed disclosure metrics.

In the agri-food sector, we conceptualize this informational lock-in as a systemic condition in which the measurement, reporting, and evaluation infrastructure used by capital markets fails to capture the material realities of a sector, thereby constraining capital allocation decisions and stabilizing incumbent, unsustainable regimes.

This process can be understood as a chain linking inputs, mechanisms, and capital allocation outcomes, all occurring under a context of the increasing pressure to act quickly under uncertainty (Figure 5). First, capital markets rely on standardized, comparable metrics to evaluate investment opportunities (Brooks et al., 2015; Gray, 2013; Mosley, 2003). However, when these metrics are applied to complex dynamics such as the ones in the agrifood systems, they fail to adequately represent material sustainability risks and opportunities (Nirino et al., 2019; van 't Klooster & Prodani, 2025; Vitale et al., 2023).

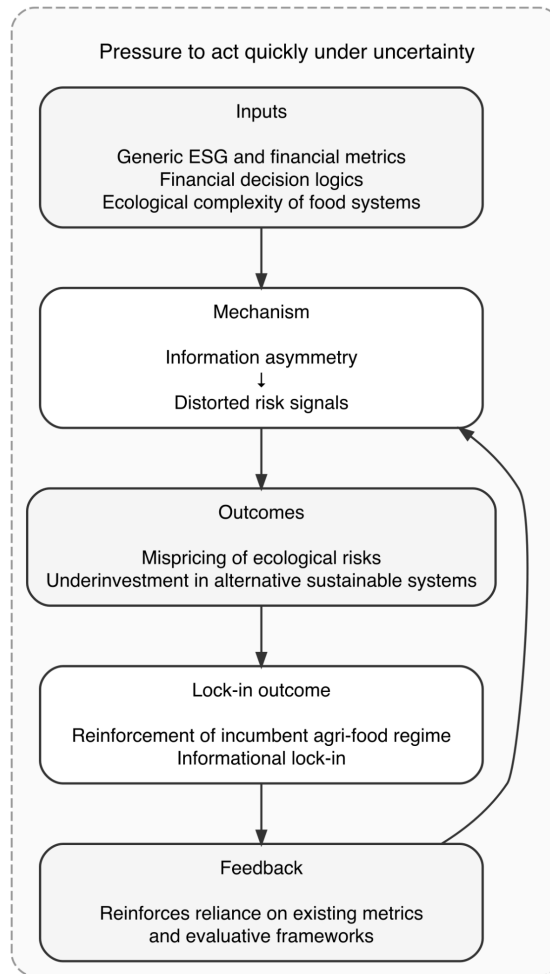


Figure 5. Figure 5. Informational lock-in dynamics

This asymmetry then creates distorted signals about risk and return, leading financial actors to misprice ecological risk, which tends to prioritize the status quo and underinvest in sustainable alternatives. However, even those willing to allocate capital to regenerative systems are hindered by the ‘material tension’: they lack the tools to see the value in biological complexity, thus leading to underinvestment in more sustainable but complex alternatives that cannot be properly captured by metrics. Over time, these allocation patterns reinforce both the dominance of incumbent agri-food regimes and the continued reliance on inadequate measurement systems, producing a self-reinforcing condition of informational lock-in. Finally, the ‘methodological tension’ blinds the academic community to this failure, as we continue to measure success using the very data points (generic ESG scores) that enforce the lock-in.

The current informational infrastructure, therefore, acts as a stabilization mechanism for the incumbent regime. It creates positive feedback loops that keep capital flowing toward industrial agriculture simply because it is legible to the financial system, while regenerative systems, which are heterogeneous and complex, remain adverse to the language of risk and so become

uninvestable in practical terms. We thus assert that the informational infrastructure itself is a binding constraint; financial actors cannot reallocate capital effectively because they cannot differentiate between genuine ecological resilience and symbolic compliance. This fact also mirrors the logic of financialization we found in the corpus in which the challenges of ecologically sustainable food systems are understood only in terms of financial risk and reward for private investors. Hence transitioning to sustainable food systems may require qualitative change in the form of alternative valuation or monitoring principles for financial actors to follow (Lagoarde-Segot, 2019; Schoenmaker & Schramade, 2019).

4.2.1 Breaking the Informational Lock-in

Breaking lock-in requires more than incremental improvements or political will, but technical and structural reform. Even if powerful financial actors wanted to reallocate capital (and evidence suggests some do) they cannot do so effectively if the institutional structures that govern its allocation remain unchanged. Conversely, structural reform of capital markets without sector-appropriate measurement infrastructure leaves financial actors unable to distinguish genuine ecological resilience from symbolic compliance. We identify three fundamental mismatches that figure research and policy must resolve.

First, the materiality mismatch. Current ESG frameworks prioritize cross-sector comparability over sector-specific relevance. As noted by Robinson et al. (2024), tools developed for other industries cannot be directly transferred to food production without losing fidelity. Future research must move beyond generic indicators to validate metrics that capture the material realities of specific food sub-sectors. Our measurement systems must reflect this heterogeneity rather than imposing standardization.

Second, temporally, financial logic is predicated on quarterly reporting and short-term liquidity, whereas ecological regeneration operates over decades. To break this cycle, research should explore metrics that capture trends in ecological outcomes (e.g., soil health trajectories) rather than static snapshots of policy commitments. This would allow capital markets to value the direction of travel, potentially reconciling the timeline conflict between financial and ecological systems.

Third, on boundaries, corporate reporting often stops at the firm's legal boundary, missing the upstream and downstream impacts that constitute the bulk of the food system's footprint. Rueda et al. (2017) provide a framework for understanding these complex supply chain interactions, yet these insights have rarely been integrated into mainstream financial analysis. A robust measurement infrastructure must extend beyond Scope 1 and 2 emissions to capture the diffuse, network-level impacts of global food supply chains.

However, the political economy perspective highlighted in this paper suggests that the challenge extends beyond simply getting the metrics right. Breaking the informational lock in must be accompanied by efforts for institutional reform within financial systems, specifically, ones which address its structural power and particular investment logics. Indeed, while there is growing

interest in alternative investment practices among select public and private financial institutions, these seem not to have penetrated into those of large multi-trillion dollar institutional investors. A political economy perspective invites finance scholars to be attendant to these power dynamics while developing alternative valuation principles.

Resolving these mismatches requires a new research agenda. To date, the dominant research question has been: does environmental disclosure correlate with financial value? But if the underlying data is flawed, positive correlations do not reflect genuine impact. Instead, the field must pivot to behavioral and infrastructural questions - how do financial actors actually interpret (or ignore) sector-specific environmental data? As Perera (2024) notes, investor responses are heterogeneous and cannot be reduced to simple binary reactions to disclosure. We need rigorous qualitative and mixed-methods research that understands investment decision-making in the agri-food sector. Only by understanding how asset managers and analysts utilize sustainability information can we design an informational infrastructure that breaks the lock-in and enables capital to flow toward genuine transformation.

4.3 Towards a third position

The two traditions we have mapped have largely operated in parallel, asking different questions of finance from different epistemological starting points. We argue that neither, in its current form, equips the field to address the practical question now confronting both scholarship and policy.

The optimization literature offers important resources, particularly its attention to the heterogeneity of food production and the inadequacy of generic financial frameworks but its working assumption that better metrics alone can redirect capital systematically underestimates the structural conditions under which information is produced, interpreted, and acted upon. The political economy on the other hand literature offers an indispensable diagnosis of those conditions. However, its analytical framing has often viewed global capital markets as detrimental to the public good (Cox, 1994; Polanyi, 1944) and has therefore tended to dismiss 'reformist reforms' in favor of 'non-reformist reforms' (Gorz, 1968), no matter how impractical or untenable they may be. The result is a literature that diagnoses extensively but rarely specifies the conditions under which the diagnosis could be acted on and seldom moves beyond theory to praxis.

Instead, we propose a third position that takes the optimistic position of the optimization literature where we can break the information lock-in so that capital markets can play a more constructive role in the transformation of food systems while remaining conscious of the structural critique. Our position rejects the dichotomy we found in the literature, which treats finance as either a neutral tool or a constitutive problem, and replaces it with a conditional question that informs a new research agenda. To date, the dominant research question has been: does environmental disclosure correlate with financial value? But if the underlying data is flawed, positive correlations do not reflect genuine impact. Instead, the field must pivot to a question whose answer can break this dichotomy: which configurations of ownership, governance, accountability, and measurement would have to hold for capital allocation decisions

to align with food system sustainability, and which actors are positioned to bring those configurations about? This agenda recognizes that the relevant agency is not confined to financial actors themselves, but civil-society actors, regulators, and producer organizations are increasingly engaging as key actors through shareholder activism, divestment coalitions, and alternative valuation frameworks rather than only resisting them externally. Only by understanding these behavioral, infrastructural, and structural conditions can we design an informational infrastructure that breaks the lock-in and enables capital to flow toward genuine transformation.

The normative posture our position calls for is similar to what Bernstein (2001) called the compromise of liberal environmentalism. A settlement in which global environmental governance has pursued ecological protection within, rather than against, liberal economic order, and which institutionalised sustainable development. In our case, we advocate a compromise that combines breaking the informational lock-in by resolving the materiality temporal and boundary mismatches that we identify, but we also go beyond Bernstein's compromise by taking seriously the longer lineage of political-economy scholarship that has shown how finance, organized under specific historical conditions, has tended to subordinate social and ecological reproduction to the logic of accumulation. The structural critique of food-system financialization we engaged in Section 3 sits within this lineage and inherits its considerable analytical power. The third position we propose retains the structural critique's attention to power, ownership, and historical specificity, while declining the foundational claim that capital markets are constitutively incapable of contributing to public goods. This is the posture we have elsewhere called *evidence-based optimism* (Clegg and Galindo-Gutiérrez, 2026), an enhanced empirical understanding of how institutions interact with capital allocation, combined with recognition that the future of finance in food-system transformation is under-determined and open to action.

5. Conclusion

This systematic review maps the academic literature at the intersection of capital markets and food systems, to ask what its structure reveals about the barriers to sustainable food finance. It challenges the prevailing narrative that the primary barrier to sustainable food systems is a lack of capital. Instead, we identify a sociotechnical clash between the short-term liquidity-driven logic of finance and the material realities of ecological systems. Despite the policy emphasis on capital markets as 'game changers', our analysis of 71 peer-reviewed articles reveals a field that is deeply fragmented and operationally disconnected from the biological complexity it tries to manage.

The scarcity of literature targeting institutional investors and capital markets highlights that while policymakers are banking on private finance, the academic understanding of these actors remains thin. We argue that the current reliance on generic toolkits has created an 'informational lock-in': a condition in which the absence of decision-useful, sector-specific metrics prevents capital reallocation even when the political will to act exists.

Consequently, the path forward is not to 'close the investment gap,' but to reconstruct the measurement infrastructure. We must move beyond the assumption that better data alone will correct market failures. Future scholarship and practice must focus on developing sector-specific logics that render the nuances of soil health, biodiversity, and nutritional density legible to capital markets.

For academia, this demands a pivot away from correlational studies of generic ESG scores, and toward behavioral investigations into how financial actors navigate, or ignore, the ecological complexity of the food system. For policymakers, this reframing matters: directing resources toward informational infrastructure may ultimately prove a more effective intervention than directing capital itself. The food system will not be transformed by capital markets that cannot distinguish between genuine ecological resilience and symbolic compliance.

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